

BCG X GenAI Simulation: Financial Data Analysis

1. Project Overview

This project is part of the BCG X Generative AI Job Simulation on Forage. The objective is to analyze the financial performance of Microsoft, Tesla, and Apple using Python and pandas.

The analysis focuses on financial trends across multiple fiscal years, including revenue, net income, total assets, total liabilities, and cash flow from operating activities.

The findings will establish a foundation for developing an AI-powered financial chatbot in the next task.

2. Project Objectives

- Analyze financial data for Microsoft, Tesla, and Apple.
- Calculate year-over-year revenue growth and net income growth.
- Compare financial performance across companies and fiscal years.
- Identify meaningful financial trends and observations.
- Document the analytical process, findings, and conclusions.
- Prepare the analysis for use in an AI-powered financial chatbot.

3. Dataset Description

The dataset contains financial information for three publicly traded companies: Microsoft, Tesla, and Apple.

The dataset covers five fiscal years, from 2021 to 2025, and contains 15 company-year records.

Financial Metrics

- **Company:** Name of the company.
- **Fiscal Year:** Financial reporting year.
- **Total Revenue:** Total income generated from business operations, reported in USD millions.
- **Net Income:** Profit remaining after expenses and taxes, reported in USD millions.
- **Total Assets:** Total resources owned or controlled by the company, reported in USD millions.
- **Total Liabilities:** Total financial obligations owed by the company, reported in USD millions.

- **Operating Cash Flow:** Cash generated from normal business operations, reported in USD millions.

Dataset Source and Limitation

This project uses a demonstration dataset for learning and practicing financial data analysis with Python and pandas. Some values may be illustrative and have not all been verified against official company filings. Therefore, the analysis is intended for educational purposes rather than as a source of verified financial advice or factual financial conclusions.

```
In [1]: import pandas as pd

df = pd.read_csv("../data/raw/BCG_X_Financial_Dataset_5_Years_DEMO.csv")

df.head()
```

```
Out[1]:
```

	Company	Fiscal Year	Total Revenue (USD millions)	Net Income (USD millions)	Total Assets (USD millions)	Total Liabilities (USD millions)	Operating Cash Flow (USD millions)
0	Microsoft	2021	168088	61271	333779	191791	76740
1	Microsoft	2022	198270	72738	364840	198298	89035
2	Microsoft	2023	211915	72361	411976	205753	87582
3	Microsoft	2024	245122	88136	512163	243686	118548
4	Microsoft	2025	281724	101832	619003	275524	136162

4. Initial Data Inspection

Before performing financial analysis, the dataset must be inspected to understand its structure, dimensions, column names, and data types.

The initial inspection helps identify potential data quality issues, missing values, and inconsistencies that could affect subsequent calculations.

The following checks will be performed:

- Display the first few records.
- Determine the number of rows and columns.
- Inspect column names and data types.
- Check for missing values.
- Review summary statistics for numerical variables.

```
In [2]: # Check the number of rows and columns
print("Dataset Shape:", df.shape)

# Display column names
print("\nColumn Names:")
```

```
print(df.columns.tolist())

# Check data types and non-null counts
print("\nDataset Information:")
df.info()

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Generate summary statistics
print("\nSummary Statistics:")
df.describe()
```

Dataset Shape: (15, 7)

Column Names:

```
['Company', 'Fiscal Year', 'Total Revenue (USD millions)', 'Net Income (USD millions)', 'Total Assets (USD millions)', 'Total Liabilities (USD millions)', 'Operating Cash Flow (USD millions)']
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 15 entries, 0 to 14
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	Company	15 non-null	object
1	Fiscal Year	15 non-null	int64
2	Total Revenue (USD millions)	15 non-null	int64
3	Net Income (USD millions)	15 non-null	int64
4	Total Assets (USD millions)	15 non-null	int64
5	Total Liabilities (USD millions)	15 non-null	int64
6	Operating Cash Flow (USD millions)	15 non-null	int64

```
dtypes: int64(6), object(1)
```

```
memory usage: 972.0+ bytes
```

Missing Values:

Company	0
Fiscal Year	0
Total Revenue (USD millions)	0
Net Income (USD millions)	0
Total Assets (USD millions)	0
Total Liabilities (USD millions)	0
Operating Cash Flow (USD millions)	0

dtype: int64

Summary Statistics:

Out[2]:

	Fiscal Year	Total Revenue (USD millions)	Net Income (USD millions)	Total Assets (USD millions)	Total Liabilities (USD millions)	Op Ca: m
count	15.00000	15.000000	15.000000	15.000000	15.000000	15.
mean	2023.00000	232021.333333	62511.666667	302149.200000	186460.066667	76246.
std	1.46385	132064.142391	41359.636246	164756.615786	112054.574182	47975.
min	2021.00000	53823.000000	3794.000000	62131.000000	30548.000000	11497.
25%	2022.00000	97231.500000	13792.000000	129535.000000	49195.000000	14835.
50%	2023.00000	211915.000000	72738.000000	352583.000000	205753.000000	89035.
75%	2024.00000	374551.000000	95837.500000	364910.000000	286456.000000	114877.
max	2025.00000	416161.000000	112010.000000	619003.000000	308030.000000	136162.



5. Data Preparation and Validation

Data preparation is an important step before performing financial analysis. It helps ensure that the dataset has an appropriate structure and that calculations can be performed reliably.

In this stage, the following checks will be performed:

- Verify that company names and fiscal years are represented correctly.
- Check for missing values and duplicate records.
- Confirm that financial metrics use numerical data types.
- Ensure that fiscal years are sorted correctly within each company.
- Review the dataset for unexpected or inconsistent values.

Because this project uses a demonstration dataset, the analysis is intended for educational purposes. Any unusual financial values will be treated as observations to investigate rather than automatically assumed to be errors.

```
In [3]: # 1. Check missing values in each column
print("Missing Values:")
print(df.isnull().sum())

# 2. Check duplicate rows
print("\nDuplicate Rows:", df.duplicated().sum())

# 3. Check data types
print("\nData Types:")
print(df.dtypes)

# 4. Check unique companies
print("\nCompanies:")
print(df["Company"].unique())
```

```
# 5. Check fiscal years
print("\nFiscal Years:")
print(sorted(df["Fiscal Year"].unique()))

# 6. Display the dataset shape
print("\nDataset Shape:", df.shape)
```

Missing Values:

Company	0
Fiscal Year	0
Total Revenue (USD millions)	0
Net Income (USD millions)	0
Total Assets (USD millions)	0
Total Liabilities (USD millions)	0
Operating Cash Flow (USD millions)	0

dtype: int64

Duplicate Rows: 0

Data Types:

Company	object
Fiscal Year	int64
Total Revenue (USD millions)	int64
Net Income (USD millions)	int64
Total Assets (USD millions)	int64
Total Liabilities (USD millions)	int64
Operating Cash Flow (USD millions)	int64

dtype: object

Companies:

['Microsoft' 'Tesla' 'Apple']

Fiscal Years:

[np.int64(2021), np.int64(2022), np.int64(2023), np.int64(2024), np.int64(2025)]

Dataset Shape: (15, 7)

6. Data Sorting and Preparation for Financial Analysis

Before calculating year-over-year financial growth, the dataset must be arranged in chronological order for each company.

Sorting ensures that each fiscal year's financial figures are compared with the preceding fiscal year's figures for the same company. This is essential for accurate growth calculations using pandas.

The dataset will be sorted by company name and fiscal year, and the index will be reset to maintain a clean sequential index.

```
In [4]: # Sort records by company and fiscal year
df = df.sort_values(
    by=["Company", "Fiscal Year"]
).reset_index(drop=True)
```

```
# Display the sorted dataset
df
```

Out[4]:

	Company	Fiscal Year	Total Revenue (USD millions)	Net Income (USD millions)	Total Assets (USD millions)	Total Liabilities (USD millions)	Operating Cash Flow (USD millions)
0	Apple	2021	365817	94680	351002	287912	104038
1	Apple	2022	394328	99803	352755	302083	122151
2	Apple	2023	383285	96995	352583	290437	110543
3	Apple	2024	391035	93736	364980	308030	118254
4	Apple	2025	416161	112010	359000	285000	111500
5	Microsoft	2021	168088	61271	333779	191791	76740
6	Microsoft	2022	198270	72738	364840	198298	89035
7	Microsoft	2023	211915	72361	411976	205753	87582
8	Microsoft	2024	245122	88136	512163	243686	118548
9	Microsoft	2025	281724	101832	619003	275524	136162
10	Tesla	2021	53823	5644	62131	30548	11497
11	Tesla	2022	81462	12587	82338	36440	14724
12	Tesla	2023	96773	14997	106618	43009	13256
13	Tesla	2024	97690	7091	122070	48390	14923
14	Tesla	2025	94827	3794	137000	50000	14747

7. Year-over-Year (YoY) Growth Analysis

Year-over-year growth measures the percentage change in a financial metric compared with the previous fiscal year.

It helps analysts understand whether a company's financial performance is improving, declining, or remaining relatively stable over time.

Revenue Growth

Revenue growth measures the percentage change in total revenue compared with the previous fiscal year.

Formula:

Revenue Growth (%) = $\frac{(\text{Current Year Revenue} - \text{Previous Year Revenue})}{\text{Previous Year Revenue}} \times 100$

Net Income Growth

Net income growth measures the percentage change in net income compared with the previous fiscal year.

Formula:
$$\text{Net Income Growth (\%)} = ((\text{Current Year Net Income} - \text{Previous Year Net Income}) / \text{Previous Year Net Income}) \times 100$$

Interpretation

- **Positive growth:** The financial metric increased compared with the previous year.
- **Negative growth:** The financial metric decreased compared with the previous year.
- **Zero growth:** The financial metric remained unchanged.
- **NaN for the first year:** Growth cannot be calculated because the previous year's data is unavailable.

Growth will be calculated separately for each company to ensure that one company's financial results are never compared with another company's results.

```
In [5]: # Calculate year-over-year revenue growth
df["Revenue Growth (%)"] = (
    df.groupby("Company")["Total Revenue (USD millions)"]
      .pct_change() * 100
)

# Calculate year-over-year net income growth
df["Net Income Growth (%)"] = (
    df.groupby("Company")["Net Income (USD millions)"]
      .pct_change() * 100
)

# Display the results
df[
    [
        "Company",
        "Fiscal Year",
        "Total Revenue (USD millions)",
        "Revenue Growth (%)",
        "Net Income (USD millions)",
        "Net Income Growth (%)"
    ]
]
```

Out[5]:

	Company	Fiscal Year	Total Revenue (USD millions)	Revenue Growth (%)	Net Income (USD millions)	Net Income Growth (%)
0	Apple	2021	365817	NaN	94680	NaN
1	Apple	2022	394328	7.793788	99803	5.410858
2	Apple	2023	383285	-2.800461	96995	-2.813543
3	Apple	2024	391035	2.021994	93736	-3.359967
4	Apple	2025	416161	6.425512	112010	19.495178
5	Microsoft	2021	168088	NaN	61271	NaN
6	Microsoft	2022	198270	17.956071	72738	18.715216
7	Microsoft	2023	211915	6.882030	72361	-0.518299
8	Microsoft	2024	245122	15.669962	88136	21.800417
9	Microsoft	2025	281724	14.932156	101832	15.539621
10	Tesla	2021	53823	NaN	5644	NaN
11	Tesla	2022	81462	51.351653	12587	123.015592
12	Tesla	2023	96773	18.795267	14997	19.146739
13	Tesla	2024	97690	0.947578	7091	-52.717210
14	Tesla	2025	94827	-2.930699	3794	-46.495558

8. Comparative Financial Performance Analysis

This section compares the financial performance of Microsoft, Tesla, and Apple over the five-year period from 2021 to 2025.

The analysis focuses on revenue growth, net income growth, and changes in profitability trends.

The comparison will help identify differences in growth patterns, highlight significant changes in financial performance, and establish a foundation for further financial analysis.

The findings are based on the demonstration dataset and should be interpreted as educational observations rather than verified financial conclusions.

```
In [6]: # Calculate average annual growth rates for each company
growth_summary = (
    df.groupby("Company")[
        ["Revenue Growth (%)", "Net Income Growth (%)"]
    ]
    .mean()
    .round(2)
)
```



```
# Display the comparative growth summary  
growth_summary
```

Out[6]:

	Revenue Growth (%)	Net Income Growth (%)
Company		
Apple	3.36	4.68
Microsoft	13.86	13.88
Tesla	17.04	10.74

9. Financial Metrics Comparison

Growth rates show how quickly financial metrics change, but they do not show the overall scale of a company's operations or financial position.

This section compares five key financial metrics across Microsoft, Tesla, and Apple:

- Total Revenue
- Net Income
- Total Assets
- Total Liabilities
- Operating Cash Flow

Average values across the available fiscal years will be calculated for each company. This comparison provides a broader view of operating performance, profitability, asset size, financial obligations, and cash generation.

All values are expressed in USD millions and are based on the demonstration dataset.

```
In [7]: # Calculate average financial metrics for each company  
financial_summary = (  
    df.groupby("Company") [  
        [  
            "Total Revenue (USD millions)",  
            "Net Income (USD millions)",  
            "Total Assets (USD millions)",  
            "Total Liabilities (USD millions)",  
            "Operating Cash Flow (USD millions)"  
        ]  
    ]  
    .mean()  
    .round(2)  
)  
  
# Display the financial comparison  
financial_summary
```

Out[7]:

	Total Revenue (USD millions)	Net Income (USD millions)	Total Assets (USD millions)	Total Liabilities (USD millions)	Operating Cash Flow (USD millions)
Company					
Apple	390125.2	99444.8	356064.0	294692.4	113297.2
Microsoft	221023.8	79267.6	448352.2	223010.4	101613.4
Tesla	84915.0	8822.6	102031.4	41677.4	13829.4

10. Financial Data Visualization

Visualizations provide a clear way to compare financial performance across companies and identify patterns that may not be immediately apparent in tables.

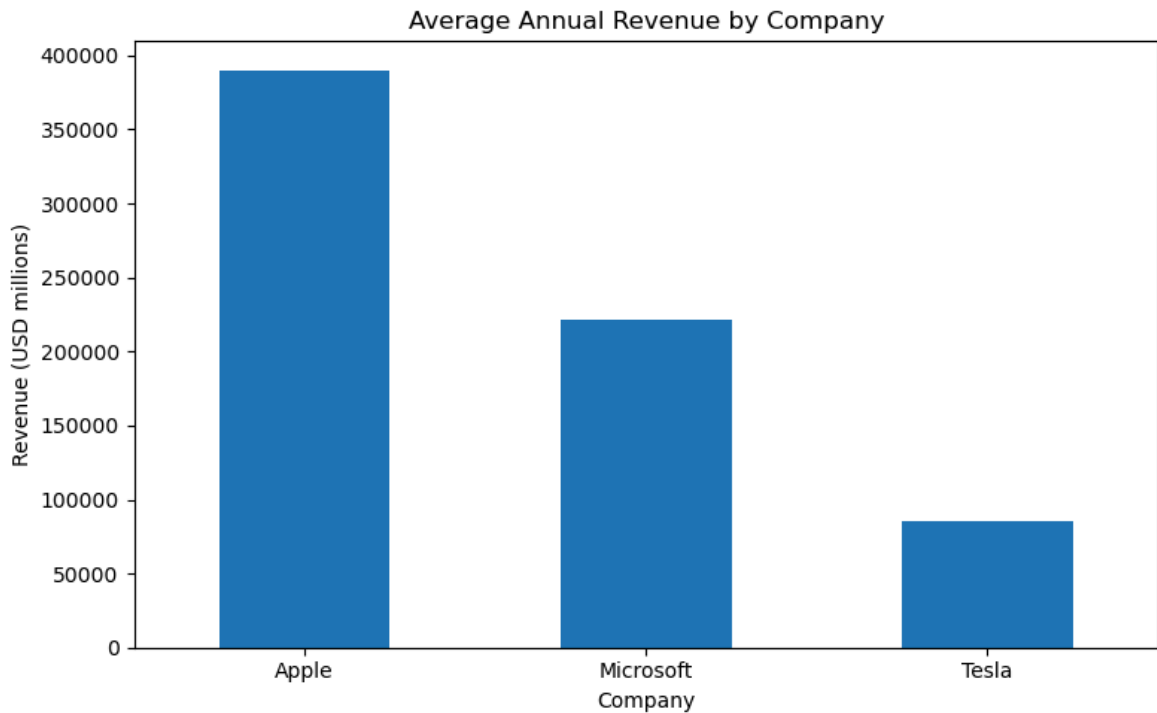
This section uses bar charts to compare the average revenue, net income, and operating cash flow of Microsoft, Tesla, and Apple.

These visualizations support the numerical analysis and help communicate the findings more effectively.

```
In [8]: import matplotlib.pyplot as plt

# Plot average revenue by company
financial_summary["Total Revenue (USD millions)"].plot(
    kind="bar",
    figsize=(8, 5),
    title="Average Annual Revenue by Company"
)

plt.xlabel("Company")
plt.ylabel("Revenue (USD millions)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
In [9]: # Create two side-by-side financial comparison charts
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

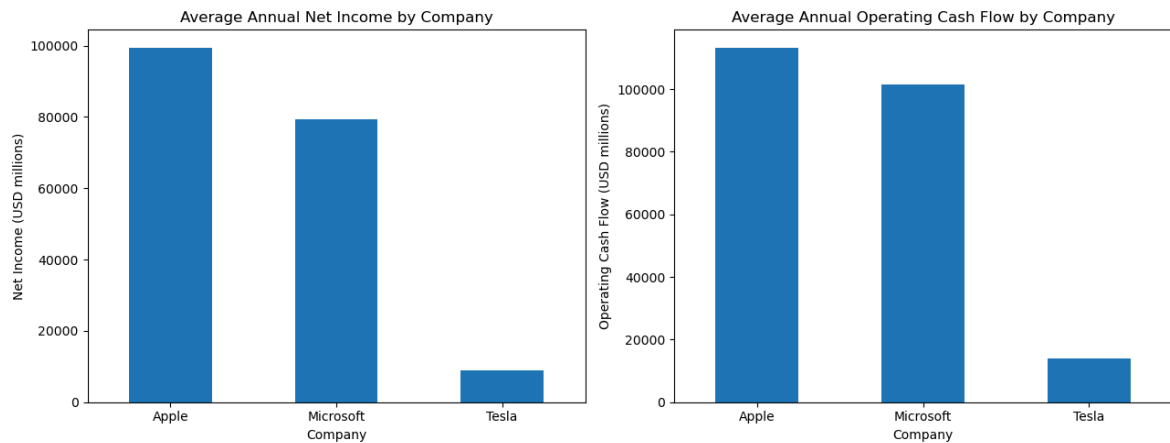
# Net income comparison
financial_summary["Net Income (USD millions)"].plot(
    kind="bar",
    ax=axes[0],
    title="Average Annual Net Income by Company"
)

axes[0].set_xlabel("Company")
axes[0].set_ylabel("Net Income (USD millions)")
axes[0].tick_params(axis="x", rotation=0)

# Operating cash flow comparison
financial_summary["Operating Cash Flow (USD millions)"].plot(
    kind="bar",
    ax=axes[1],
    title="Average Annual Operating Cash Flow by Company"
)

axes[1].set_xlabel("Company")
axes[1].set_ylabel("Operating Cash Flow (USD millions)")
axes[1].tick_params(axis="x", rotation=0)

plt.tight_layout()
plt.show()
```



11. Key Findings and Observations

Based on the demonstration dataset covering 2021–2025, the following observations were identified.

1. Revenue Growth

Tesla recorded the highest average annual revenue growth at 17.04%, followed by Microsoft at 13.86% and Apple at 3.36%.

2. Net Income Growth

Microsoft recorded average annual net income growth of 13.88%, followed by Tesla at 10.74% and Apple at 4.68%.

3. Revenue Scale

Apple had the highest average annual revenue at USD 390,125.2 million, followed by Microsoft at USD 221,023.8 million and Tesla at USD 84,915 million.

4. Profitability

Microsoft recorded the highest average net income at USD 79,267.6 million, followed by Apple at USD 99,444.8 million and Tesla at USD 8,822.6 million.

5. Operating Cash Flow

Apple recorded the highest average operating cash flow at USD 113,297.2 million, followed by Microsoft at USD 101,613.4 million and Tesla at USD 13,829.4 million.

Important Limitation

These observations are based on a demonstration dataset created for educational purposes. They should not be interpreted as verified financial statements or investment recommendations.

12. Conclusion

This project demonstrated how Python and pandas can be used to inspect, validate, analyze, and visualize financial data.

The analysis included data quality checks, year-over-year revenue and net income growth calculations, comparative financial summaries, and visualizations of key financial metrics.

The results highlighted differences in revenue scale, profitability, growth patterns, and operating cash flow among the three companies in the demonstration dataset.

This analysis provides a foundation for the next stage of the project: developing an AI-powered financial chatbot that can answer financial questions using structured company data.

All findings are educational and depend on the illustrative values contained in the dataset.

In []: